

Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B. Tech.****Branch : Petrochemical Engineering****Semester :VI****Subject Code & Name: BTPCC603 PROCESS DYNAMICS AND CONTROL****Max Marks: 60****Date:23/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Laplace transform of a linear or ramp forcing function having input of slope B is given as- a. BS b. B/S^2 c. BS^2 d. S^2B	Level 1	1
2	A resistance which has the linear relationship between flow and head is known as a. Non- linear. b. Linear. c. Exponential. d. Additional	Level 1	1
3	The phase lag for first order system subjected to sinusoidal forcing function can never exceed ---- a. 45° b. 60° c. 70° d. 90°	Level 1,2	1
4	The time constant for thermometer is given by a. $\frac{MC}{hA}$ b. $(MC)*(hA)$ c. hA/MC d. hA	Level 1,2	1
5	A pure capacitive process is- a. self-regulating. b. non - self-regulating c. fast. d. transfer lag.	Level 1	1
6	The term interacting is generally referred to as ---- a. loading. b. unloading. c. empty. d. capacity.	Level 1	1
7	The control configuration in which the computer is used to schedule the operation including changes in demand, supply and prices with time is known as--- a. Scheduling computer control b. Adaptive control c. Feed forward control d. DDC	Level 1	1
8	The control configuration in which the optimization of the plant's economic performance is performed using programmed intelligence of a computer is known as---- a. supervisory computer. control. b. feedforward control. c. feedback control d. ratio control	Level 1	1
9	A control system is unstable at the phase lag 180° for which the open loop frequency response exhibits an amplitude ratio exceeding	Level 1,2	1

	a. one.	b. two.	c. three.	d. zero.			
10	The corner frequency for PD controller having derivative time τ_D is given by				Level 1	1	
	a. τ_D .	b. τ_D^2 .	c. τ_D^4 .	d. $\frac{1}{\tau_D}$.			
11	In most of the open-loop transfer functions for single-loop chemical process control systems the open-loop poles are on ---				Level 1	1	
	a. imaginary axis.	b. equidistant from the imaginary axis.	c. closed loop transfer function.	d. real axis.			
12	Asymptotes are the straight lines radiating from the				Level 1	1	
	a. center of gravity.	b. open-loop pole.	c. Break in point.	d. Breakaway point.			
Q. 2	Solve the following.					12	
A)	Obtain y(t)				Level 1,2,3	6	
	$\frac{s + 1}{s^2 + 2s + 5}$						
B)	With appropriate example discuss the principal characteristics of the first order systems and describe the physical meaning of the time constant for the first order system.				Level 1, 2	6	
Q.3	Solve the following.					12	
A)	In a process consist of non-interacting two tank multicapacity control system in series. The inflow q would be considered as manipulated variable and the liquid head h_2 in the lower tank would be considered as the variable to be controlled. Inflows q_1 and q_2 considered as load variables. Downstream head at the lower tank is considered constant. h_1 is the head in the upper tank and q_r and q_0 are the outflows at the upper and the lower tanks respectively. Assume the flow resistance to the linear. Determine the operational equation for the overall system relating q, q_1 , q_2 and q_0 with the time constants t_1 and t_2 of the respective tanks.				Level 2, 3, 4	6	
B)	The overall transfer function of control system is $G(s) = \frac{16}{1.5s^2+2.4s+6}$. A step change of magnitude 6 is introduced into the system. Determine (1) Overshoot, (2) Period of oscillation, (3) ultimate value of response				Level 2, 3	6	
Q. 4	Solve Any Two of the following.					12	
A)	Give brief account of different modes of controllers used in process control.				Level 1,2	6	
B)	Draw the root-locus diagram for the system whose transfer function is given by				Level 2,3,4	6	
	$G(s) H(s) = \frac{K(s+1)}{s^2(s+2)} \quad 0 < K < \infty$						
C)	The characteristic equation of a closed loop control system is $s^4 + 3s^3 + 5s^2 + 4s + 2 = 0$ Determine the stability by the Routh criterion.				Level 2,3,4	6	
Q.5	Solve Any Two of the following.					12	
A)	Plot the Bode diagram for the following transfer function				Level 1,2	6	
	$\frac{10}{s(s+1)(s+5)}$						

B)	Give the informative account of graphical rules for Bode Diagrams.	Level 2,3,4	6
C)	Give the informative account of the following. Controller Selection and Tuning Feed forward Control	Level 2,3,4	6
Q. 6	Solve Any Two of the following.		12
A)	Give informative account of the following electronic and pneumatic transmitters used in the industries.	Level 1,2	6
B)	With neat sketch discuss sight glass level indicator used in fluid level measurement in the industries	Level 1,2	6
C)	With neat sketch discuss bimetallic thermometer.	Level 2,3,4	6
	*** End ***		